SU-GAB

ATOM COUNT CARD • MOLE CONCEPT&MOLECONVERSIONS • Chemistry for Engineers • Activity 02

B. BASE QUANTITATIVE DATA



A. PRODUCT IDENTITY

Su-gab White Sugar

Net weight: 500 g

Compound: Sucrose

Formula: $C_{12}H_{22}O_{11}$

Role: Sweetener / energy

Real label photo shown on the left ←

Mass of sucrose	500 g (treated as 3 s.f.)
Formula	$C_{12}H_{22}O_{11}$
Atomic masses	C = 12.011 H = 1.008 O = 15.999g/mol
Avogadro constant	6.02214076×10^{23} mol^{-1} (exact)
Scale-up scenario	50 packs (exact count)

C. MOLAR MASS BUILD-UP

El	Atoms	Ar	Contrib.	% mass
C	12	12.011	144.132	42.107%
H	22	1.008	22.176	6.479%
O	11	15.999	175.989	51.414%
Σ	—	—	342.297	100%

M = 12(12.011)+22(1.008)+11(15.999) = 342.297 g/mol

D. THE MOLE MAP (one 500 g pack)

500 g sucrose —(÷ M, 342.297)→ **1.46 mol** sucrose —(× N_A, exact)→ **8.80×10^{23}** molecules

Label bridge: Net weight = mass of pure sucrose

Mass → moles: 500 g × (1 mol / 342.297 g) = 1.4607 → 1.46 mol

Moles → particles: 1.4607 mol × 6.02214076×10^{23} = 8.80×10^{23} molecules

E. INSIDE THE FORMULA

Carbon atoms in one pack

8.7967×10^{23} molecules × 12 C atoms = 1.06×10^{25} C atoms

% Composition + two-route check (Carbon)

%C = [12×12.011 / 342.297] × 100% = 42.107%

Route A: 500 g × 0.42107 = 210.535 g C

Route B: 1.4607 mol × 12 × 12.011 = 210.536 g C → both 211 g C ✓

Reverse → empirical formula

No common factor in subscripts → empirical = molecular formula (n = 1) ✓

F. SCALE-UP & REVERSE

Scale-up to 50 packs

50 × 1.4607 mol = 73.0 mol sucrose

Mass = 2.50×10^4 g | Molecules = 4.40×10^{25}

Reverse: How many packs = 1.00 mol?

1.00 mol × 342.297 g/mol ÷ 500 g/pack = 0.685 pack

≈ two-thirds of one pack contains one mole of sucrose

G. SIG-FIG CHECK & ONE-NUMBER TAKEAWAY

- 500 g treated as 3 significant figures (5.00×10^2 g)
- N_A, subscripts and pack counts are exact (do not limit)
- Calculator value 8.7967×10^{23} → reported 8.80×10^{23} molecules

ONE-NUMBER TAKEAWAY

One 500 g pack holds 8.80×10^{23} sucrose molecules (~1.46 mol).

One mole of sucrose has a mass of only about 342 g.

H. REFERENCES & AI DISCLOSURE

https://www.pnphome.co.za/huletts-white-sugar-500g/p/000000000000208422_EA

National Center for Biotechnology Information. (n.d.). PubChem compound summary for CID 5988: Sucrose. PubChem. https://pubchem.ncbi.nlm.nih.gov/compound/Sucrose

Commission on Isotopic Abundances and Atomic Weights. (n.d.). Standard atomic weights. https://ciaaw.org/atomic-weights.htm

AIdisclosure:AI was used to organize the layout. All label values, formulas and final answers were independently checked by the student.